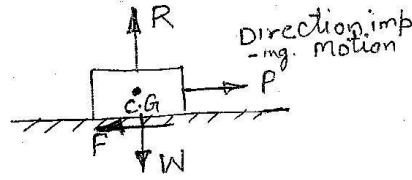


Friction; and Frictional force: When a body moves over another body, a opposing force is exerted in between two surfaces in contact, this opposing or resisting force is known as force of friction and the property by virtue of which this force exerted is known as friction.

→ Consider a body resting on a horizontal plane as shown in fig.



Let W = Weight of body acting vert. downward through C.G. of the body.

R = Normal reaction of body acting vertically upward through the C.G. of body

P = Force applied on the body also through the C.G. and $\parallel$ to H.P.

→ Frictional force is a self adjusting force up to some instant. As we apply a small force on the body, the body will not move as the force of friction balance the applied force. If we start increase the applied force gradually, F will also increase and a stage will come when the motion of body is about to begin, at that stage the magnitude of frictional force will be known as limiting force of friction which is the maximum value of frictional force.

We can define the static and kinetic friction with the help of the fig.

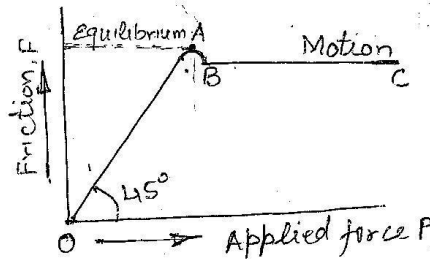
→ At Pt A the force of friction will be limiting force of friction.

→ The frictional force from pt O to just before pt A is known as static friction i.e. when the body is at rest

→ The frictional force from point B onward

is known as dynamic friction i.e. when the body is in motion which is also known

as kinetic friction: Dynamic friction may be dry friction or fluid friction.



- ⇒ Dry friction: The friction between dry surfaces in contact is called dry friction. It is also known as solid or Coulomb friction. Dynamic friction is further sub-divided as
- (a) Sliding friction: Friction between two surfaces when one surface slides over another surface.
 - (b) Rolling friction: Friction between two surfaces which are separated by balls or rollers, i.e. when one body rolls over the surface of another body.
 - It may be noted that rolling friction is always less than sliding friction.

* The another type of friction is fluid friction and defined as the friction between two surfaces in the presence of fluid.

⇒ Laws of dry friction: Based upon the experimental observations, following laws of friction have been established for dry or solid contact surfaces.

- (1) Limiting friction or total friction is independent of the area and shape of contact surfaces.
- (2) Friction acts tangential to the surfaces in contact and in opposite direction to that in which motion is to take place.
- (3) Frictional force is maximum at the instant of impending motion. It varies from zero to limiting friction depends upon resultant force tending to cause motion.
- (4) Limiting friction depends upon the roughness of the surfaces in contact.
- (5) The magnitude of limiting friction bears a constant ratio to the normal reaction between the contact surfaces.
- (6) At low velocities between sliding surfaces, the frictional force is practically independent of the velocity. But the force necessary to impending motion is greater than that which is necessary to maintain the motion.

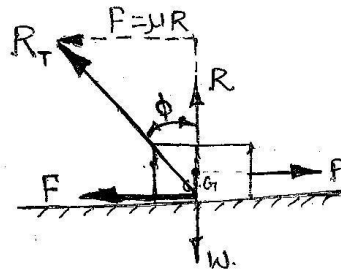
⇒ Co-efficient of friction: It is defined as the ratio of limiting force of friction to the normal reaction between two bodies. It is denoted by μ . Mathematically

$$\mu = \frac{\text{Limiting force of friction}}{\text{Normal reaction}} = \frac{F}{R} \quad \text{or } F = \mu R$$

⇒ Angle of Friction: It is defined as the angle which the resultant of limiting force of friction and normal reaction makes with the normal reaction and denoted by ϕ as shown in fig.

Let R_T is the resultant of limiting force of friction (F) and Normal reaction (R)

Then ϕ is the angle of friction angle between R_T & R

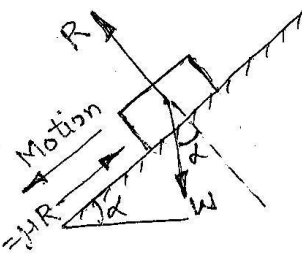


Then $\tan \phi = \frac{F}{R} = \frac{\mu R}{R} = \mu$ which is co-efficient of friction

That is why we can say tangent of angle of friction is equal to the co-efficient of friction.

⇒ Angle of repose: Consider a block resting on an inclined plane making an angle λ with the horizontal. If the angle λ is increased gradually till the block is just at the point of sliding downward at that instant the inclined angle λ is said to be angle of repose, as shown in fig.

Now the block may be considered in equilibrium under the action of three forces. Wt of block = W acting vertically down word, Normal reaction R vertically up words and limiting frictional force $F = \mu R$ upward along the plane.



Then by resolving the forces along the plane & $\perp$ to the plane
 $\mu R = W \sin \lambda$ and $R = W \cos \lambda \Rightarrow \mu = \tan \lambda$.

In terms of angle of friction $\mu = \tan \phi$

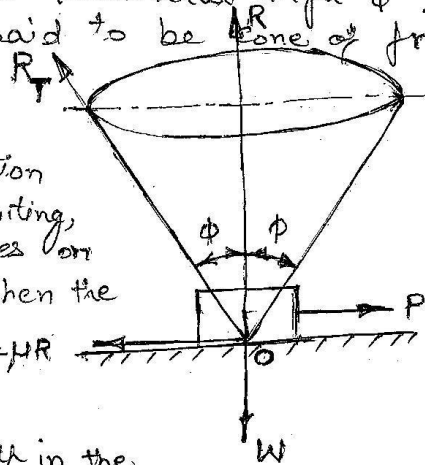
⇒ $\tan \lambda = \tan \phi \Rightarrow \lambda = \phi \Rightarrow \text{angle of repose} = \text{angle of friction}$

⇒ Cone of friction: As shown in fig if the applied force P is given a complete revolution about the normal reaction R then the total reaction R_T generate a right circular cone having vertex O , axis R and semi-vertical angle ϕ . Such a right circular cone is said to be cone of friction.

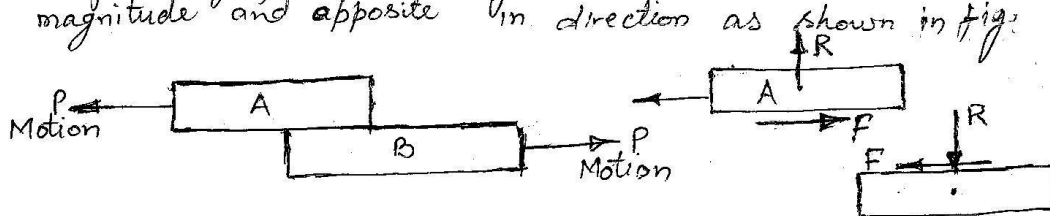
It is clear from the fig. that in case of impending motion the frictional force is limiting, and the total reaction R_T lies on the surface of cone. When the

frictional force is less $F = \mu R$ than the limiting friction, the total reaction would lie within the

cone. In this case the force acting on the body is not sufficient to cause motion. This aspect forms the working principle for self-locking mechanisms.



⇒ When two bodies are in contact and both have a tendency to move w.r.t. each other, then the normal reaction as well as force of friction acting on two bodies are equal in magnitude and opposite in direction as shown in fig.

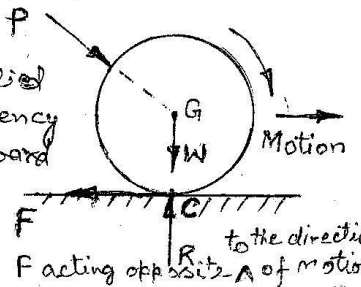


⇒ Force of friction on a wheel or cylinder, sphere or disc.

The above said bodies can be brought in motion by

(1) Applying an external force (2) Applying a torque at the centre.

⇒ Case I When an external force is applied: When an external force P is applied to the wheel as shown in fig., the tendency of applied force is to move the wheel in forward direction and the frictional force F opposes the forward translatory motion by acting at the contact pt. C from right to left.

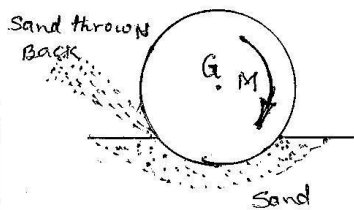


The frictional force F shall produce a turning moment equal to $F \times R$ about the C.G. of the wheel thus giving the clockwise rolling motion to the wheel. It ^{shall} depend upon the magnitude of friction present whether the wheel will slide, roll or both slide as well as roll.

If the wheel rolls only without sliding then the frictional force F present is less than μR . The equation $F = \mu R$ is valid only when the wheel is at the Pt. of sliding or slipping.

Case 2. When acted upon by a Torque: This situation may be represented if the wheel is power driven as in case of an automobile wheel. Imagine the wheels of a car sunk in the sand as shown in fig. 1. At this stage the wheel is acted upon by a torque from the engine shall keep spinning back the sand, without moving forward.

In this way we observe that in the absence of friction the wheel keeps rotating without moving forward. The wheel only be able to move in forward direction in the presence of sufficient friction! Thus when the wheel is powered, the basic tendency is to rotate the wheel and it is the friction only that provide the necessary forward force to give the forward motion to the wheel. Thus friction in this case acts in the ~~same~~ direction of motion of the wheel, as shown in fig. 2.



The wheel just spins only in the absence of friction

Fig. 1

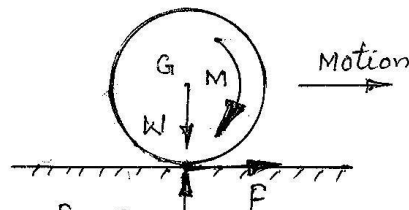
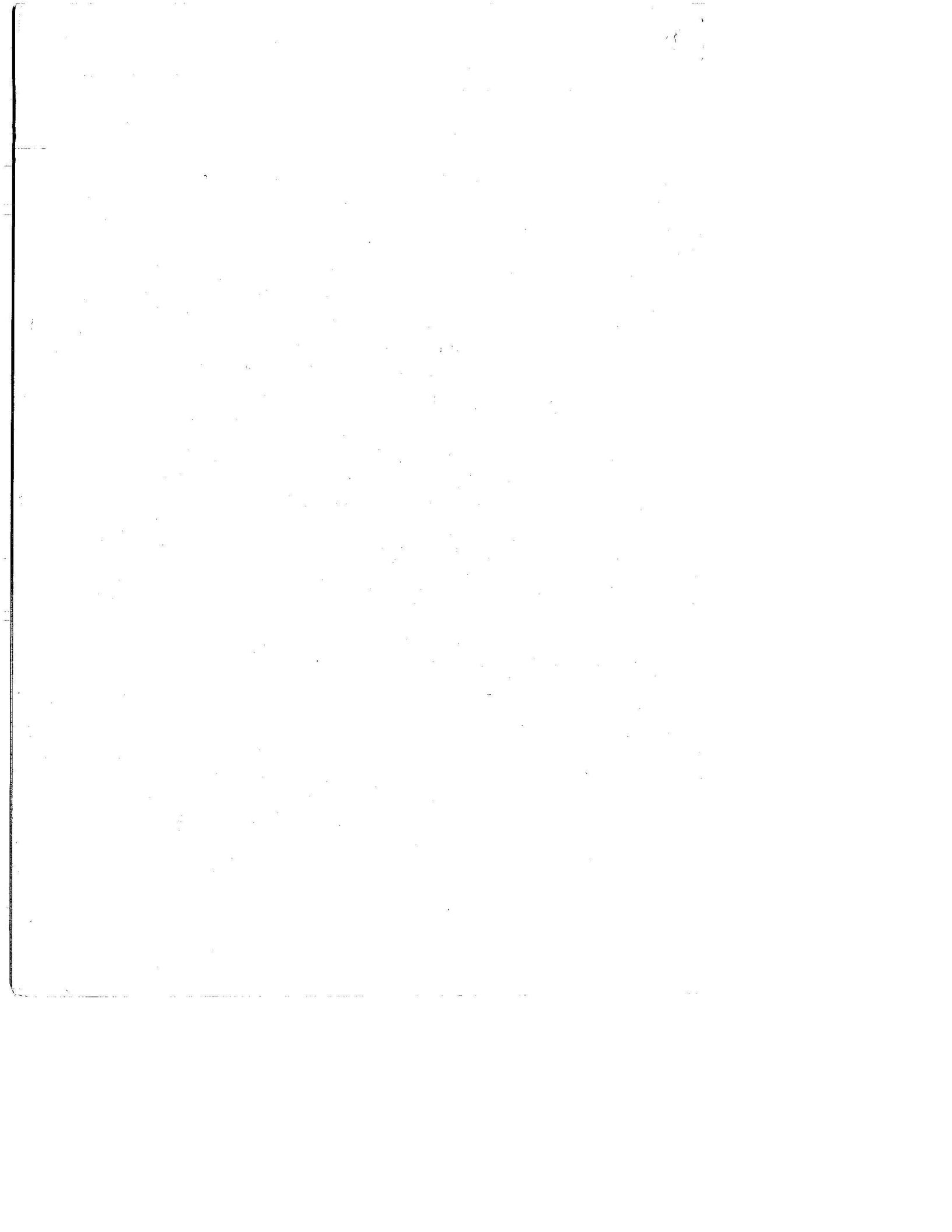


Fig. 2.

Frictional force ' F ' makes the wheel move in forward direction. In this case frictional force ' F ' is acting in the direction of motion of wheel.



Ladder Friction: A ladder is a device which is used for climbing on walls or roofs. It consists of two long up-rights of wood, iron, aluminium or any other material connected by a no. of cross pieces. These cross pieces are known as rungs and provide step for climbing.

Consider a ladder resting on the ground at point A and leaning against a wall ^{at Pt. B} as shown in fig.

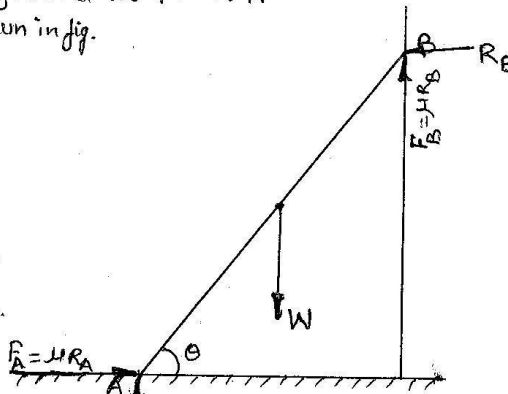
Let R_A = reaction at Point A

R_B = reaction at Point B

F_A = force of friction at A = μR_A

F_B = force of friction at B = μR_B

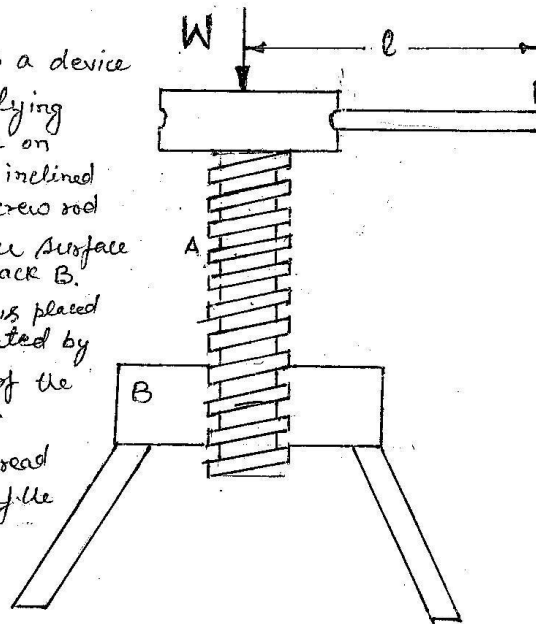
W = Wt of the ladder acting vert. downwards through G.C.



Due to the self wt of ladder or when a man stands on the ladder, the upper end B of the ladder tends to slip downwards and the force of friction between the ladder and the vertical wall $F_B = \mu R_B$ will be act upwards. Similarly the lower end A of the ladder tends to move toward the left of the wall and the force of friction $F_A = \mu R_A$ will act toward the wall.

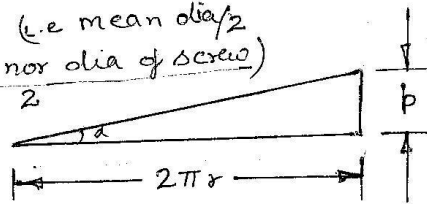
For Solving the Problems on ladder friction, consider the ladder in equilibrium under the action of all the forces and apply the Conditions of equilibrium. i.e. $\sum F_x = 0$, $\sum F_y = 0$, $\sum M = 0$

Screw Friction: The screw jack is a device used for lifting heavy load, by applying small effort at its handle. The principle on which screw jack works is similar to an inclined plane. It consists of a threaded rod A called screw rod. The screw has square threads, on its outer surface which fits into the inner threads of the jack B. The load required to be raised or lowered is placed on the head of the screw, which is rotated by the application of an effort at the end of the lever for lifting or lowering the load.



If one complete revolution of the screw thread be imagined to be unwound from the body of the screw and developed it will form an inclined plane of base $2\pi r$ equal to p , pitch of the screw as shown in fig. on next page.

Let p = pitch of the screw (i.e. the distance between two consecutive threads)
 r = mean radius of the screw (i.e. mean dia/2)
 mean dia = $\frac{\text{major dia} + \text{minor dia of screw}}{2}$
 α = Helix angle



Then from the given fig.

$$\tan \alpha = \frac{p}{2\pi r} \quad \text{in single start threaded screw}$$

$$\tan \alpha = \frac{np}{2\pi r} \quad \text{in multi-start threaded screw}$$

where n is the no. of threads in one lead

Now Let. P_r = effort applied at the mean radius of the screw jack to lift the load.

W = weight of the body to be lifted

μ = Co-efficient of friction between the screw & nut

ϕ = Angle of friction such that $\mu = \tan \phi$

P_l = effort applied at the end of the lever
 where l is the length of the lever

As the working principle of screw jack is similar to the principle of Inclined plane, so the effort applied can be calculated in the similar manner.

That is why

The effort required to lift the load by screw jack is given by

When effort is applied at mean radius $P_r = W \tan(\alpha + \phi)$
 and effort applied at the ^{end of} lever of length l

$$P_l = \frac{P_r \times r}{l}$$

And the effort required to lower the load

When applied at mean radius $P_r = W \tan(\phi - \alpha)$
 (when $\alpha > \phi$)

or

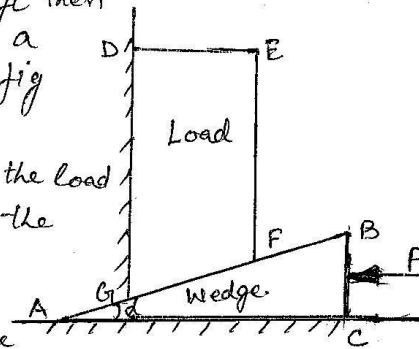
$$P_r = W \tan(\phi - \alpha)$$

The effort applied at the end of lever of length l (When $\phi > \alpha$)

$$P_l = \frac{P_r \times r}{l}$$

Wedge Friction: A wedge is a piece of metal or wood which is generally of triangular or trapezoidal in cross-section. It is used either for lifting loads or for slight adjustment in the position of body i.e. for tightening fits or keys in case of shafts.

When a heavy load is required to be left then the wedge is placed below the load and a horizontal force is applied as shown in fig



As if the force P is just sufficient to lift the load the wedge will move towards left and the load will move upward. When the wedge moves towards left the sliding of surfaces

AC & AB will take place. At the same time load moves upwards and sliding of the load takes place along GD . Thus for the wedge and load shown in fig sliding takes place along surfaces AB , AC and DG . Therefore three Normal reactions will act at the surfaces AB , AC & DG

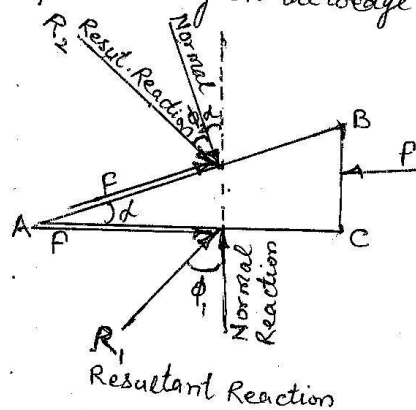
The problems on wedges are basically the problems of equilibrium on inclined planes. Thus these problems may be solved either by the conditions of equilibrium or by applying Lam's theorem

Solution by Equilibrium conditions: In this method the equilibrium of wedge and the equilibrium of load are considered separately.

NOTE: If load is given then consider the equilibrium of load firstly and if force P is given then consider the equilibrium of wedge firstly & load secondly.

1. Consider the equilibrium of wedge firstly. The forces acting on the wedge are shown in fig i.e. F.B.D. These are

- (i) The force P applied horizontally on surface BC
- (ii) Reaction R_1 on surface AC (R_1 is the resultant of normal reaction and force of friction on surface AC , which will be inclined at an angle ϕ_1 with the normal, ϕ_1 is angle of friction,
- (iii) R_2 on the surface AB , which is the resultant of Normal reaction on surface AB & force of friction on surface AB , R_2 will be inclined at an angle ϕ_2 with the normal.



When force P is applied on the wedge, the surface CA will move toward left and hence force of friction will act toward right. In the same way the force of friction on AB face will act from A to B as shown in F.B.D of wedge.

Now Resolving the forces horizontally we have,

$$R_1 \sin \phi_1 + R_2 \sin(\phi_2 + \alpha) = P$$

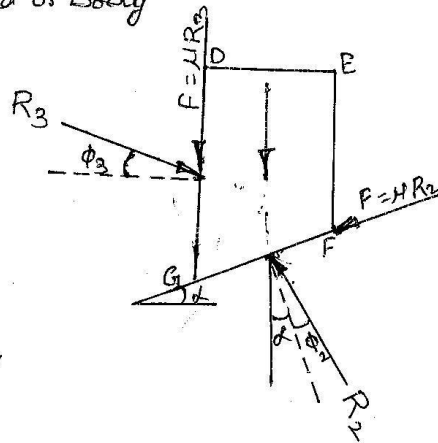
Resolving the forces vertically we have

$$R_1 \cos \phi_1 = R_2 \cos(\phi_2 + \alpha)$$

2. Now consider the Equilibrium of load or Body

The forces acting on the body are shown in F.B.D of Body. These are

- (i) Wt W of the Body
- (ii) Reaction R_3 on the surface GD which is the resultant of normal reaction and force of friction on surface GD .
- (iii) R_2 on surface GE .



Now Resolving the forces horizontally

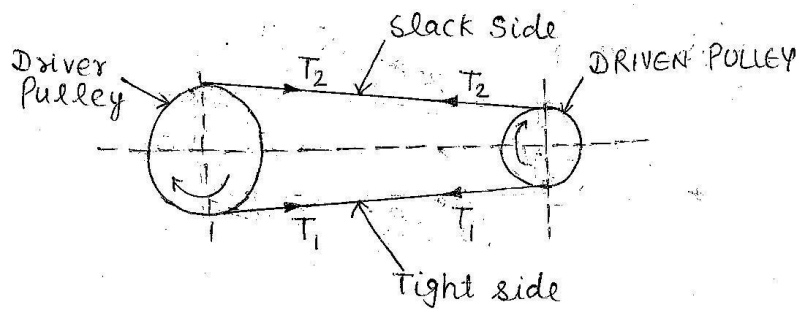
$$R_3 \cos \phi_3 = R_2 \sin(\phi_2 + \alpha)$$

Resolving the forces vertically we have

$$W + R_3 \sin \phi_3 = R_2 \cos(\phi_2 + \alpha)$$

In this way we will get the required solution of problems on the wedges.

RATIO OF BELT TENSIONS :-



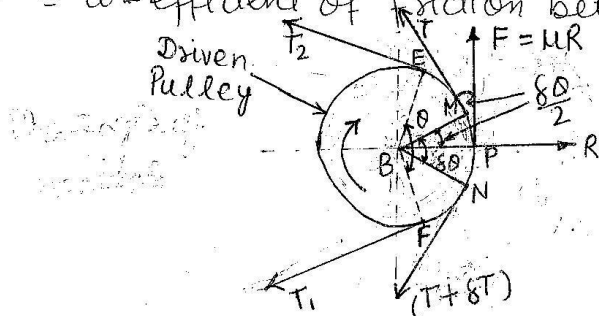
Above figure shows a driver pulley A and driven pulley B rotating in the clockwise dirⁿ. Consider the driven pulley B.

Let T_1 = Tension in the belt on the tight side

T_2 = Tension in the belt on the slack side

θ = Angle of Contact i.e., the angle subtended by the arc EF at the centre of the driven pul

μ = co-efficient of friction between belt and pulley.



The ratio of the two tensions may be found by considering an elemental piece of belt MN subtending an angle $\delta\theta$ at the centre of the pulley B. The various forces which keep the elemental piece MN in equilibrium are:

- i) Tension T in the belt at M acting tangentially,
- ii) Tension $T + \delta T$ in the belt at N acting tangentially,
- iii) Normal rxn. R acting radially outward at P, where P is the middle point of MN,
- iv) Frictional Force $F = \mu R$ acting at right angles to R and in the opposite dirⁿ of the motion of Pulley.

Now angle $PBM = \frac{\delta\theta}{2}$. Also angle $TPF = \frac{\delta\theta}{2}$

Resolving all the forces acting on the belt MN in the horizontal dirn, we get

$$R = T \sin \frac{\delta\theta}{2} + (T + \delta T) \sin \frac{\delta\theta}{2}$$

Since the angle $\delta\theta$ is very small, $\sin \frac{\delta\theta}{2}$ can be written as $\frac{\delta\theta}{2}$. Hence the above eqn becomes as

$$\begin{aligned} R &= T \times \frac{\delta\theta}{2} + (T + \delta T) \times \frac{\delta\theta}{2} \\ &= T \times \frac{\delta\theta}{2} + T \times \frac{\delta\theta}{2} + \delta T \times \frac{\delta\theta}{2} \\ &= T \times \delta\theta + \frac{\delta T \times \delta\theta}{2} = \underline{T \times \delta\theta} \quad \text{--- (i)} \end{aligned}$$

(Neglecting small quantity $\frac{\delta T \times \delta\theta}{2}$)

Now resolving all the forces vertically, we get $\rightarrow$

$$F = (T + \delta T) \cos \frac{\delta\theta}{2} - T \cos \frac{\delta\theta}{2}$$

Since $\delta\theta$ is very small, hence $\cos \frac{\delta\theta}{2}$ reduces to unity i.e. 1. Hence the above eqn becomes as

$$F = (T + \delta T) - T = \delta T$$

$$\mu R = \delta T \quad (\because F = \mu R)$$

$$R = \frac{\delta T}{\mu} \quad \text{--- (ii)}$$

Equating the two values of R given by eqn (i) and (ii), we get

$$T \times \delta\theta = \frac{\delta T}{\mu} \Rightarrow \frac{\delta T}{T} = \mu \cdot \delta\theta$$

Integrating the above equation between limits T_2 & T_1 we get.

$$\int_{T_2}^{T_1} \frac{dT}{T} = \int \mu \cdot d\theta = \mu \int d\theta$$

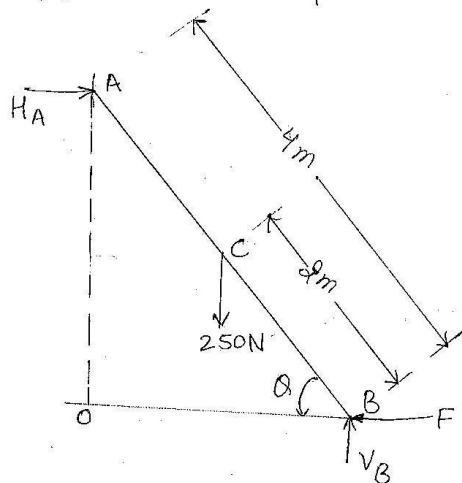
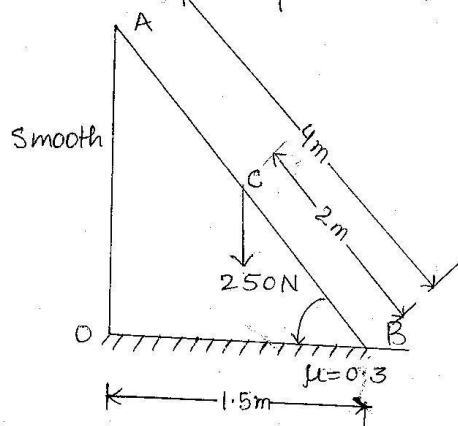
$$\log_e \frac{T_1}{T_2} = \mu \times \theta$$

$$\boxed{\frac{T_1}{T_2} = e^{\mu \times \theta}}$$

θ should be taken in radians. Here θ is known as angle of contact.

Ladder Friction :-

Q-1. A 4m ladder weighing 250 N is placed against a smooth vertical wall with its lower end 1.5 m away from the wall. If the coefficient of friction between the ladder and the floor is 0.3, show that the ladder will remain in equilibrium in this position.



$$\cos \theta = \frac{1.5}{4} \text{ and } \sin \theta = \frac{3.7}{4} \text{ Since } OA = 3.7 \text{ m}$$

$$\cos \theta = 0.375$$

$$\sin \theta = 0.925$$

For the ladder to be in equilibrium, let F be the required frictional force at the foot of the ladder.

$$\sum F_x = 0 \quad H_A - F = 0$$

$$\sum F_y = 0 \quad V_B - 250 = 0$$

$$V_B = 250$$

$$\sum M \text{ about } B = 0 \quad H_A \times (4 \sin \theta) - 250 \times 2 \cos \theta = 0$$

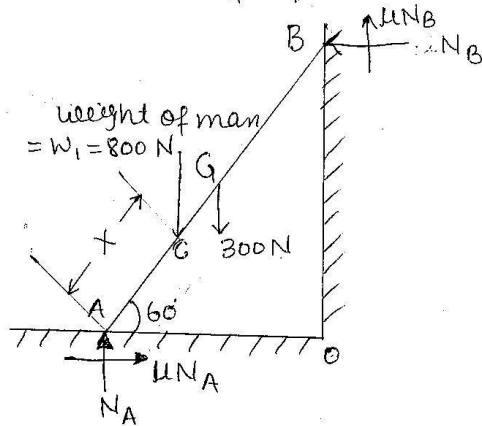
$$H_A = \frac{250 \times 2 \times 0.375}{4 \times 0.925} = 50.67 \text{ N}, \quad H_A = 50.67 \text{ N}$$

But available frictional force at $B = 0.3 V_B = 0.3 \times 250$

Since the available friction force is more than the required frictional force at B , the ladder

will remain in equilibrium.

Q-2 A 6m long uniform ladder weighing 300 N is placed against a vertical wall. Its inclination with the ground is 60° . At what distance from the lower end of the ladder a man weighing 800 N should climb before slipping of ladder starts. The coefficient of friction between contact surfaces is 0.2



Solⁿ length of ladder = $AB = 6\text{ m}$, $G = \text{CG of ladder}$
 $AG = GB = 3\text{ m}$, weight of ladder = $300\text{ N} = W$
 weight of man = $W_1 = 800\text{ N}$
 Let the man reach C before slip occurs.

$$\text{By } \sum F_y = 0 \quad N_A + \mu N_B - 300 - 800 = 0$$

$$N_A + 0.2 N_B = 1100 \quad \text{--- (1)}$$

$$\text{By } \sum F_x = 0 \quad \mu N_A - N_B = 0$$

$$N_B = \mu N_A = 0.2 N_A \quad \text{--- (2)}$$

By substituting value of N_B from eq. (2) into eq. (1) we get $\rightarrow$

$$N_A + 0.2 (0.2 N_A) = 1100$$

$$N_A = \frac{1100}{1.04} = 1057.69\text{ N}$$

and $\sum M_A = 0$ and $N_B = 0.2 N_A = (0.2)(1057.69) = 211.54\text{ N}$

$$W_1 \times AC_1 + WAG_1 - N_B \times BO - \mu N_B \times AO = 0$$

$$800 \times x \cos 60 + 300 \times 3 \cos 60 - 211.54 \times 6 \sin 60 - 0.2 \times 211.54 \times 6 \cos 60 = 0$$

$$400x + 450 - 1099.2 - 126.9 = 0$$

$$\Rightarrow \boxed{x = 1.94\text{ m}}$$

a) The ladder is in equilibrium $\therefore$

$$\text{By } \sum F_y = 0$$

$$N_A + \mu_A N_B - W_L - W_m = 0$$

$$N_A + 0.3 N_B - 800 - 200 = 0$$

$$N_A + 0.3 N_B = 1000 \quad \text{--- (1)}$$

$$\text{By } \sum F_x = 0$$

$$\mu_A N_A + P - N_B = 0$$

$$P = N_B - \mu_A N_A$$

$$P = N_B - 0.4 N_A \quad \text{--- (2)}$$

$$\text{By } \sum M_A = 0$$

$$N_B \times OB + \mu_B N_B \times OA - W_m \times OA - W_L \times AG_1 = 0$$

$$N_B \times 4 \sin 50 + 0.3 N_B \times 4 \cos 50 - 800 \times 4 \cos 50 - 200 \times 2 \cos 50 = 0$$

$$\therefore 3.836 \times N_B = 2314.035$$

$$\therefore N_B = 603.24 \text{ N}$$

$$\therefore N_A = 1000 - 0.3 N_B$$

$$N_A = 819.03 \text{ N}$$

$$\text{From (2)} \quad P = 603.24 - 0.4 \times 819.03$$

$$P = 275.628 \text{ N}$$

b) when force P is not applied let the ladder have inclination θ with the floor.

Since the ladder is in equilibrium

$$\text{By } \sum F_y = 0 \text{ we get: } \rightarrow$$

$$N_A + \mu_A N_B - W_L - W_m = 0$$

$$N_A + 0.3 N_B - 200 - 800 = 0$$

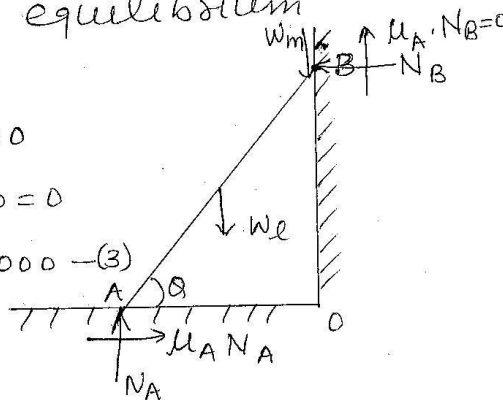
$$N_A + 0.3 N_B = 1000 \quad \text{--- (3)}$$

$$\text{By } \sum F_x = 0$$

$$\mu_A N_A - N_B = 0$$

$$N_B = \mu_A N_A$$

$$N_B = 0.4 N_A \quad \text{--- (4)}$$

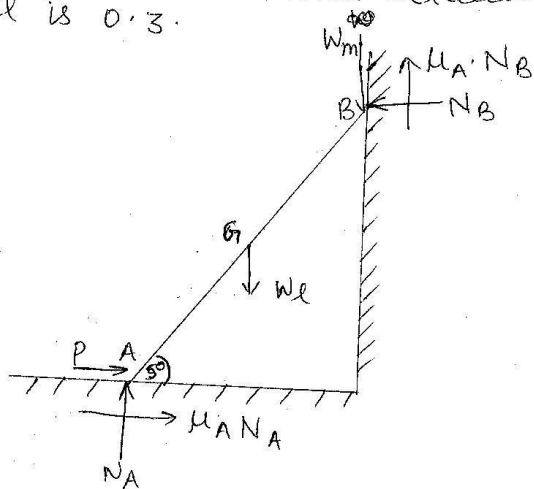


Q-3 A uniform ladder 4 m long weighs 200 N. It is placed against a wall making an angle of 50° with the floor.

In addition to its own weight, the ladder supports a man 800 N at its top. Calculate

- The horizontal force to be applied, to the ladder at its end resting on the floor to prevent slipping.
- If the force is not applied find the inclination of the ladder with the floor so that it does not slip when the man is at its top.

The coefficient of friction between the ladder and floor is 0.4 and between the ladder and the wall is 0.3.



Solⁿ

Let W_L = weight of ladder = 200 N

W_m = weight of man = 800 N

$AB = 4\text{ m}$ = length of ladder

$AG = BG = 2\text{ m}$

Man is at B and the G is the C.G. of ladder.

Let P = force applied at A to prevent slipping of ladder.

Eqⁿ (3) gives

$$N_A + 0.3 \times (0.4 N_A) = 1000$$

$$N_A = \frac{1000}{1.12} = 892.86 \text{ N}$$

and from Eqⁿ (4)

$$N_B = 0.4 \times 892.86 = 357.14 \text{ N}$$

By $\Sigma M_A = 0$, we have

$$N_B \times OB + 0.3 N_B \times OA - W_m \times OA - W_L \times AG_1 = 0$$

$$357.14 \times (4 \times \sin \alpha) + 0.3 \times 357.14 \times (4 \cos \alpha) - 800 \times 4 \times \cos \alpha - 200 \times 2 \cos \alpha =$$

$$1428.56 \sin \alpha - 3171.43 \cos \alpha = 0$$

$$\tan \alpha = \frac{3171.43}{1428.56} = 2.22$$

$$\alpha = \tan^{-1} 2.22 = 65.75^\circ$$

Ques. 4

The pitch of a screw jack is 12 mm, the mean diameter of the square thread on the spindle is 60 mm, and the lever is 900 mm long.

If $\mu_s = 0.08$, find the necessary effort at the end of the lever $\rightarrow$

- for lifting a load of 30 kN
- for lowering a load of 30 kN
- efficiency of the screw jack for raising the load.
- efficiency of the screw jack for lowering the load.

Soln

Given : Pitch = 12 mm
mean diameter, $d = 60 \text{ mm}$
 $\mu_s = 0.08$
length of lever = 900 mm
Weight, $W = 30 \text{ kN}$

$$\therefore \tan \alpha = \frac{P}{\pi d} \Rightarrow \alpha = \tan^{-1} \left(\frac{P}{\pi d} \right) = \tan^{-1} \left(\frac{12}{60\pi} \right) = 3.64^\circ$$

$$\phi = \tan^{-1} \mu_s = \tan^{-1} 0.08 = 4.57^\circ$$

$$\phi + \alpha = 4.57^\circ + 3.64^\circ = 8.21^\circ \text{ and } (\phi - \alpha) = 0.93^\circ$$

a) Effort on the surface of spindle ^{for raising the load} $= P_{r+} = W \tan(\phi + \alpha)$
 $P_{r+} = (30 \times 1000) \times \tan 8.21^\circ = 4328.4 \text{ N}$ for raising the load
Effort at the end of lever $= P_{l+}$

$$P_{l+} = \frac{P_{r+} \times d}{2l} = \frac{4328.4 \times 60}{2 \times 900} = 144.28 \text{ N}$$

b) Effort on the surface of spindle for lowering the load $= P_{r-} = W \tan(\phi - \alpha)$

$$\therefore P_{r-} = (30,000) \tan 0.93^\circ = 486.99 \text{ N}$$

$\therefore$ Effort at the end of lever to lower the load $= P_{l-} = \frac{P_{r-} \times d}{2l} = \frac{486.99 \times 60}{2 \times 900} \text{ N}$

$$P_{l-} = 16.23 \text{ N}$$

c) Ideal effort $= P_0 = W \tan \alpha = 30,000 \times \tan 3.64^\circ = 1908.4 \text{ N}$

$\therefore$ Efficiency for raising the load $= \eta_{+}$

$$\eta_{+} = \frac{P_0}{P_{r+}} \times 100 = \frac{1908.47}{4328.4} \times 100 = 44.09\%$$

d) Efficiency for lowering the load

$$= \eta_{-} = \frac{P_{r-}}{P_0} \times 100 = \frac{486.99}{1908.47} \times 100 = 25.52\%$$

Ans

Screw Jack

Q-1. The Pitch of a screw Jack is 12 mm, the mean diameter of the square thread on the spindle is 60 mm and the lever is 900 mm long. If $\mu_s = 0.08$, find the necessary effort at the end of lever :-

- for lifting a load of 30 kN
- for lowering a load of 30 kN
- efficiency of the screw Jack for raising the load
- efficiency of the screw jack for lowering the load.

Solⁿ Given :- Pitch, $p = 12 \text{ mm}$
mean diameter, $d = 60 \text{ mm}$
 $\mu_s = 0.08$
length of lever, $l = 900 \text{ mm}$
weight, $W = 30 \text{ kN}$

$$\therefore \tan \alpha = \frac{p}{\pi d} \Rightarrow \alpha = \tan^{-1}\left(\frac{p}{\pi d}\right) = \tan^{-1}\left(\frac{12}{60\pi}\right)$$
$$= \tan^{-1}(0.06366)$$

$$\therefore \alpha = 3.64^\circ \text{ and } \phi = \tan^{-1} 0.08 = 4.57^\circ$$

$$\therefore \alpha + \phi = 3.64^\circ + 4.57^\circ = 8.21^\circ \text{ and } \phi - \alpha = (4.57^\circ - 3.64^\circ)$$
$$= 0.93^\circ$$

a) Effort on the surface of spindle = $P_r = W \tan(\alpha + \phi)$

$$\therefore P_r = (30 \times 1000) \times \tan 8.21^\circ = 4328.4 \text{ N for raising the load}$$

$$\text{Effort at the end of lever to raise the load} = P_e = \frac{P_r \times d}{2l}$$
$$= \frac{4328.4 \times 60}{2 \times 900} = 144.281$$

b) Effort on the surface of spindle for lowering the load = $P_r = W \tan(\phi - \alpha)$

$$\therefore P_r = (30,000) \tan 0.93^\circ = 486.99 \text{ N}$$

$\therefore$ Effort at the end of lever to lower the load

$$= P_{e-} = \frac{P_r \times d}{2l} = \frac{486.99 \times 60}{2 \times 900} \text{ N}$$

$$\boxed{P_{e-} = 16.23 \text{ N}}$$

c) ideal effort = $P_o = W \tan \alpha = 30,000 \times \tan 3.64^\circ$
 $= 1908.47 \text{ N}$

$\therefore$ Efficiency for raising the load = η_{raising}

$$\eta_+ = \frac{P_o}{P_{r+}} \times 100$$
$$= \frac{1908.47}{4328.4} \times 100$$

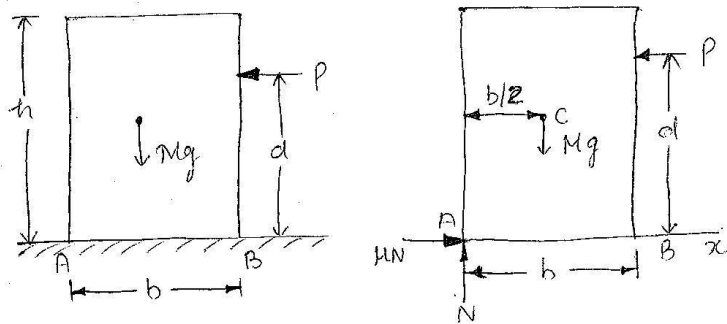
$$= 44.09\%$$

d) efficiency for lowering the load = η_{lowering}

$$\eta_- = \frac{P_{r-}}{P_o} \times 100$$
$$= \frac{486.99}{1908.47} \times 100$$

$$= 25.52\%$$

- Q. A rectangular block of mass M rests on a floor. The coefficient of friction b/w the block & the floor is μ . what is the highest position for a horizontal force P that would permit it to just move the block without tipping.



Solⁿ As tipping occurs, the block will just be lifted off the floor & shall maintain a contact with the floor only at the edge A. So, the normal reaction N & the frictional force μN shall act at A. Writing the equation of equilibrium

$$\sum F_x = 0 \quad \mu N - P = 0 \quad \text{--- (i)}$$

$$\sum F_y = 0 \quad N - Mg = 0 \quad \text{--- (ii)}$$

Taking moment about A

$$\sum M_A = 0 \quad P(d) - Mg\left(\frac{b}{2}\right) = 0 \quad \text{--- (iii)}$$

From (i) & (ii) $N = Mg$ & $P = \mu Mg$

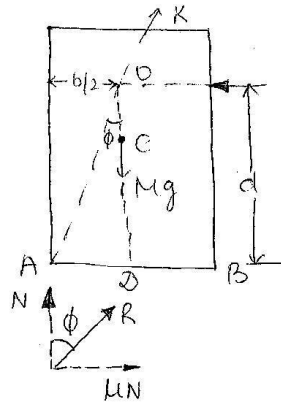
Substituting in (iii)

$$\mu Mg d = Mg \frac{b}{2}$$

$$d = \frac{b}{2\mu}$$

The problem can also be solved by combining the normal reaction N & the frictional force μN into a single resultant reaction R .

Then, the forces acting on the block reduce to a system of three concurrent forces in equilibrium.



ϕ = angle of friction

$$\tan \phi = \mu$$

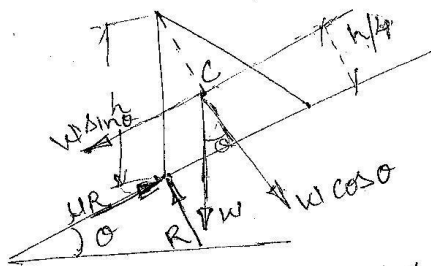
From ΔAOD ,

$$\tan \phi = \frac{AD}{OD} = \frac{b/2}{d}$$

$$\mu = \frac{b}{2d}$$

$$d = \frac{b}{2\mu} \text{ Ans}$$

A Right cone of height h & the base of radius r rests on an inclined plane as shown in fig. The co-efficient of friction between the plane and cone is μ . If the angle of inclination of the plane is slowly increased till the cone is just at the point of toppling over find the relation between μ , r & h .



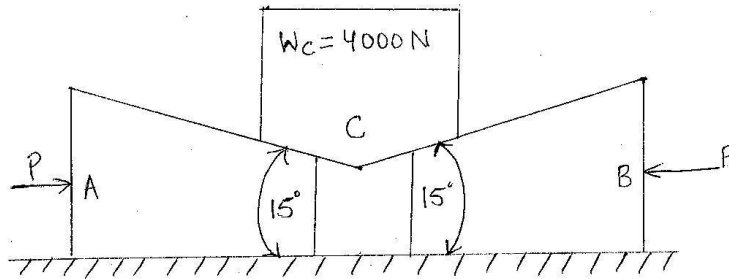
Forces acting are shown in fig under which the cone may be considered in equilibrium. When the cone will ^{at the pt of} toppling, the reaction will ^{act} at the corner.

For finding out relation between r , μ & h take moment of all the forces about centre of Gravity

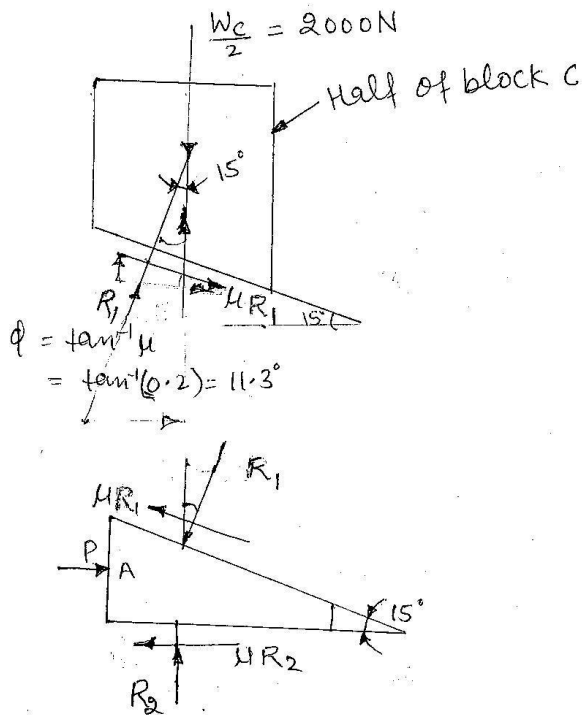
$$\sum M_C = \mu R \times h/4 = R \times r \Rightarrow \mu \times h/4 = r \text{ or } \mu = \frac{4r}{h}$$

8. Two wedges A and B are used to raise a 4000 N block C by applying horizontal force P as shown in the following figure.

Find the effort P when $\mu = 0.20$ for each surface in contact.



Solⁿ Since both wedges are similar therefore it is better to consider half of the weight of block C ($= \frac{1}{2} W_c = 2000 \text{ N}$) to be raised by each wedge. Let us consider the FBDs of half of the block and one wedge as in following figure.



For block C

By $\Sigma F_y = 0$, we get

$$R_1 \cos 15^\circ - \mu R_1 \sin 15^\circ - 2000 = 0 \quad \text{--- (1)}$$

$$\therefore R_1 (\cos 15^\circ - 0.2 \sin 15^\circ) = 2000$$

$$\therefore R_1 = \frac{2000}{0.9142} = 2187.71 \text{ N Ans}$$

For the wedge

By $\Sigma F_y = 0$, we get

$$R_2 - R_1 \cos 15^\circ + \mu R_1 \sin 15^\circ = 0$$

$$\therefore R_2 = R_1 (\cos 15^\circ - 0.2 \sin 15^\circ)$$

$$R_2 = 2187.71 \times (0.9142)$$

$$R_2 = 1999.92 \text{ N}$$

And by $\Sigma F_x = 0$, we get

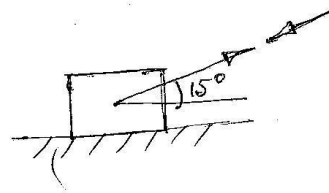
$$P - \mu R_2 - R_1 \sin 15^\circ - \mu R_1 \cos 15^\circ = 0$$

$$\therefore P - 0.2 \times 1999.92 - 2187.71 \times \sin 15^\circ - 0.2 \times 2187.71 \cos 15^\circ = 0$$

$$P = 1394.2 \text{ N Ans}$$

Determine the force required to ^{just} pull the block to right and push it to left. If the wt of the block is 100N and $\mu = 0.3$. The force is making an angle of 15° to the horizontal plane as shown in fig.

Let P_1 be the pull required to just pull the block. F.B.D of block is shown



Applying the conditions of Equilibrium.

$$\sum F_x = 0 \quad F = P_1 \cos \theta$$

$$\sum F_y = 0 \quad R = W - P_1 \sin \theta$$

$$\text{But } F = \mu R \therefore F = \mu(W - P_1 \sin \theta) = \mu W - \mu P_1 \sin \theta$$

$$\therefore P_1 \cos \theta = \mu W - \mu P_1 \sin \theta$$

$$\text{or } P_1 \cos \theta + \mu P_1 \sin \theta = \mu W$$

$$\text{or } P_1 = \frac{\mu W}{\cos \theta + \mu \sin \theta} = \frac{0.3 \times 100}{\cos 15^\circ + 0.3 \times \sin 15^\circ} = \frac{30}{0.966 + 0.3 \times 0.259} = \frac{30}{0.966 + 0.0777}$$

$$P_1 = \frac{30}{0.966 + 0.0777} = \frac{30}{1.0436} = 28.74 \text{ N}$$

Let P_2 be the Push required.

Applying Conditions of Equilibrium. $F = \mu R$

$$\sum F_x = 0 \quad F = P_2 \cos \theta$$

$$\sum F_y = 0 \quad R = W + P_2 \sin \theta$$

$$F = \mu R, F = \mu(W + P_2 \sin \theta) = \mu W + \mu P_2 \sin \theta$$

$$\therefore P_2 \cos \theta = \mu W + \mu P_2 \sin \theta \quad \text{or } P_2 \cos \theta - \mu P_2 \sin \theta = \mu W$$

$$\therefore P_2 = \frac{\mu W}{\cos \theta - \mu \sin \theta} = \frac{0.3 \times 100}{\cos 15^\circ - 0.3 \times \sin 15^\circ} = \frac{30}{0.966 - 0.3 \times 0.259} = \frac{30}{0.966 - 0.0777}$$

$$P_2 = \underline{33.77 \text{ N}}$$

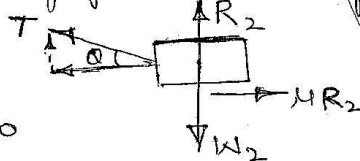
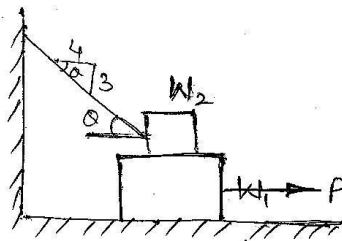
6.5. A block of wt $W_1 = 1000 \text{ N}$ rests on a horizontal surface, and supports on its top another block of wt $W_2 = 250 \text{ N}$. The wt W_2 is attached by an inclined string AB to the vertical wall. Find the magnitude of the horizontal force P applied to the lower block to cause slipping to impend. The coefficient of friction for all surfaces may be assumed to $\mu = 0.2$.

From the given data

$$\tan \theta = \frac{3}{4} = .75 \quad \theta = 36.86^\circ$$

Let T be the tension in string

Considering the Equilibrium of W_2 and applying the conditions of Equilibrium



$$\sum F_x = 0$$

$$T \cos \theta = \mu R_2 \quad \text{or} \quad T \cos 36.86 = .3 R_2 \quad \text{or} \quad .8 T = .3 R_2 \quad \text{or} \quad R_2 = \frac{0.8}{0.3} T$$

$$\sum F_y = 0 \quad R_2 = W_2 - T \sin \theta$$

$$R_2 = 2.66 T$$

$$\text{Now } 2.66 T = 250 - T \sin 36.86$$

$$2.66 T = 250 - 0.6 T$$

$$3.26 T = 250$$

$$T = \frac{250}{3.26} = 76.69 \text{ N}$$

$$R_2 = 250 - 76.69 \times \sin 36.86 = 250 - 46 = 204 \text{ N}$$

$$\mu R_2 = 204 \times 0.2 = 40.8$$

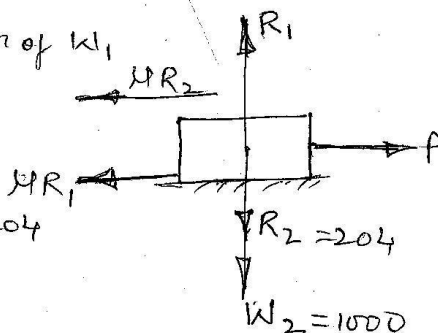
Considering the equilibrium of W_1

$$\sum F_x = 0$$

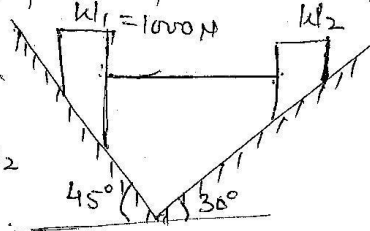
$$P = \mu R_1 + \mu R_2$$

$$R_1 = W_1 + R_2 = 1000 + 204 = 1204$$

$$P = .3 \times 1204 + 40.8 = 422.4$$



Two blocks of load $W_1 = 1000 \text{ N}$ and W_2 are resting on two rough inclined planes are connected by a horizontal link as shown in fig. Find the value of Ranges of wt W_2 (minimum & maximum value of wt W_2) for which the equilibrium can exist. take the Co-efficient of friction for both the planes is 0.364.



Minimum value of W_2

For the minimum value of wt W_2 the wt W_2 will be at the point of sliding upward hence the W_1 will be at the point of sliding down ward.

$\therefore$ The force required to keep the wt W_1 in equilibrium is given by: $P_1 = W_1 (\tan(\alpha - \phi)) = 1000 \times \tan(45 - 20) \text{ N}$

$$P_1 = 1000 \times 0.4663 = 466.3 \text{ N}$$

The same force will keep the wt W_2 in equilibrium i.e. to prevent the wt W_2 from motion up the plane

$$\therefore P_1 = W_2 (\tan(\alpha + \phi))$$

$$\therefore 466.3 = W_2 \tan(30 + 20) = W_2 \tan 50 = 1.1917 W_2$$

$$\text{or } W_2 = 466.3 / 1.1917 = 391.27 \text{ N}$$

For Maximum value of W_2

For max. value of wt W_2 , the wt W_2 will move motion down the plane & the wt W_1 will move motion up the plane same force is required to keep the blocks in equilibrium. Let it be P_2

Considering the Equilibrium of block W_1

$$P_2 = W_1 (\tan(\alpha + \phi)) = 1000 \times \tan(45 + 20) = 2144.5 \text{ N}$$

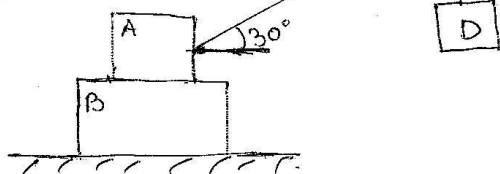
$P_2 = 2144.5 \text{ N}$ will also keep the wt W_2 in equilibrium down the plane from sliding

$$\therefore P_2 = W_2 \tan(\alpha - \phi)$$

$$\therefore 2144.5 = W_2 \tan 10^\circ = 0.1763 W_2$$

$$W_2 = \frac{2144.5}{0.1763} = 12163.9 \text{ N}$$

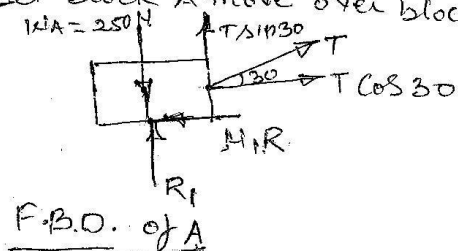
$W_A = 250 \text{ N}$
 $W_B = 500 \text{ N}$
 μ_1 between A & B = 0.4
 μ_2 between B & plane = 0.1
 μ_3 between pulley & belt = 0.3



Wt of D = W_D
 Determine the least wt of block D for motion of the block D to impend.

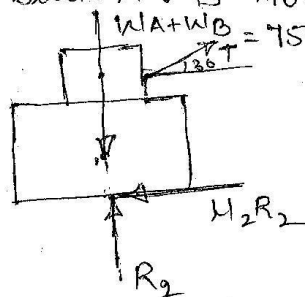
There are two Possibilities (I) Block A move over block B

(I) Let block A move over block B.



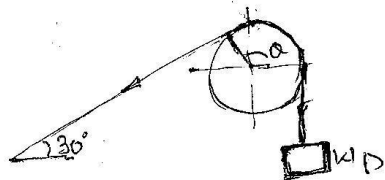
$$\begin{aligned}
 \sum F_x = 0 \text{ gives} \\
 T \cos 30 &= \mu_1 R_1 \text{ or } 0.866T = 0.4R_1 \\
 \sum F_y = 0, 250 &= R_1 + T \sin 30 \\
 250 &= 0.2165T + 0.5T = 0.7165T \\
 T &= \frac{250}{0.7165} = 348.91 \text{ N}
 \end{aligned}$$

(II) Block A & B move Together



$$\begin{aligned}
 \sum F_x = 0 \text{ gives} \\
 T \cos 30 &= \mu_2 R_2 \text{ or } 0.866T = 0.1R_2 \\
 0.866T &= 0.1R_2 \\
 T &= \frac{0.1}{0.866} R_2 = 0.1154 R_2 \\
 \sum F_y = 0 \text{ gives} \\
 750 &= R_2 + T \sin 30 \\
 750 &= R_2 + 0.5T \\
 750 &= 0.866T + 0.5T \quad T = \frac{750}{1.366} = 549.78 \text{ N}
 \end{aligned}$$

$$T_{\min} = 549.78 \text{ N}$$



T_1 Tight side tension in the belt

$$\text{Using } \frac{T_1}{T_2} = e^{\mu \theta}$$

$$\frac{W_D}{81.87} = e^{0.3(120 \times \frac{\pi}{180})} = e^{1.884}$$

$$\frac{W_D}{81.87} = 2.718^{1.628} = 1.874$$

$$W_D = 1.874 \times 81.87 = 153.45 \text{ N}$$

The angle θ sub. by the belt at the centre of pulley = $90 + 30 = 120$
 Let slack side Tension $T_2 = T_{\min} = 81.87 \text{ N}$
 For Least W_D $W_D \leq T_{\min}$

3.2 Definitions:

original sequences (as shown in the map of locations in Figure 1d). [5]

the original mapping of IAS in T). As we will discuss later in the paper, this case does not
 point privacy threats. When selecting locations to truncate from sequences, our method
 considers both the satisfaction of the privacy constraints and also the introduced error
 $= 2.852$
 $R_2 = 2 + 0.852$
 $P = 1.3 \times 0.852 + 0.3 \times 2.852$
 each respect to the original trajectory. For example, the deleted points in the transformed
 points close to their neighboring points in the corresponding

The first trajectory in T_0 is mapped to four trajectories $t_01 - t_04$ in which b_1 and b_2 appear with probability 50% - Pba. The same holds for the last three trajectories and b_2 appear with probability 50% - Pba. Note that t_05 cannot be mapped to any trajectory in T_0 . In the worst case, the adversary can infer that t_05 must be one of t_05 to t_08 , since the A-projection a_3 of these trajectories appears only three times in T_A (and a_1 could have been truncated from T_A).

For each adversary A and for every projected sequence $TA \in TA$ (corresponding to $t \in T$), we would like the adversary, not to be able to reconstruct any point in $p \in \Pi$, if $p \in TA$.
 $1 = 2.8867 + 5T$ or $1 = 3.3667$ or $T = 1/3.8867$
 with certainty higher than pbr . For example, if $pbr = 50\%$, then the exact publication of

$$\sum F_y = 0 \quad 1 = R_1 + T \sin 30^\circ \text{ or } 1 = R_1 + .5T$$

measured by the similarity of the original trajectories in T to their corresponding

...the information are (i) the satisfaction of certain privacy requirements

1. The two communicating

...suppression of location information in T wherever

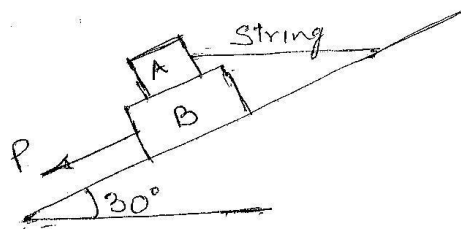
7.8m30+ from using their partial knowledge (linked to a person's identity) to infer

the method of publishing the trajectories is studied in a way that would prevent

Find Force P required to move the

$W_A = 1 \text{ kN}$ $W_2 = 2 \text{ kN}$ $W_1 \text{ per } a = 0.8$

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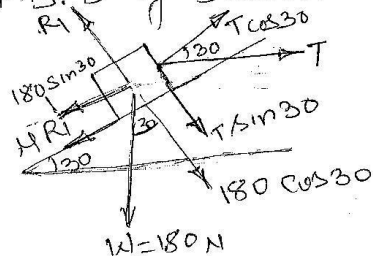
$$W_A = 180 \text{ N} \quad W_B = 270 \text{ N}$$

$$\mu \text{ for all surfaces} = 0.4$$

String is horizontal

Find the tension in String
& Pull P applied Parallel to
the plane, required for motion
down the Block B.

F.B.D of Block A



$$\sum F_x = 0 \text{ Gives}$$

$$T \cos 30 - 180 \sin 30 - \mu R_1 = 0$$

$$T \cos 30 = 180 \sin 30 + 0.4 R_1$$

$$1.866 T = 90 + 0.4 R_1 \quad \text{--- (I)}$$

$$\sum F_y = 0 \text{ Gives}$$

$$T \sin 30 + 180 \cos 30 = R_1$$

$$0.5 T + 155.88 = R_1 \quad \text{--- (II)}$$

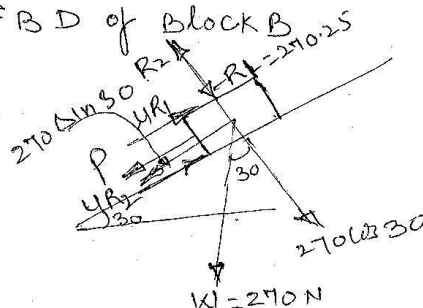
$$0.866 T = 90 + 0.4 (0.5 T + 155.88)$$

$$0.866 T = 90 + 0.2 T + 62.35$$

$$0.666 T = 152.35 \Rightarrow T = \frac{152.35}{0.666} = 228.75 \text{ N}$$

$$R_1 = 0.5 \times 228.75 + 155.88 = 270.25 \text{ N}$$

F.B.D of Block B



$$\sum F_x = 0 \text{ Gives}$$

$$\mu R_1 + \mu R_2 = P + 270 \sin 30$$

$$0.4 (R_1 + R_2) = P + 135$$

$$0.4 (270.25 + R_2) = P + 135$$

$$\sum F_y = 0 \text{ Gives}$$

$$270 \cos 30 + R_1 = R_2$$

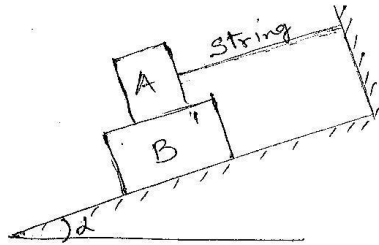
$$233.82 + 270.25 = R_2 \Rightarrow R_2 = 504.07$$

$$\text{So } P + 135 = 0.4 (270.25 + 504.07)$$

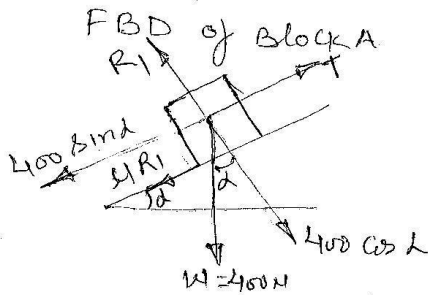
$$P + 135 = 0.4 (774.32)$$

$$P = 0.4 (774.32) - 135 =$$

$$P = 309.73 - 135 = \underline{\underline{174.74 \text{ N}}}$$



$W_A = 400\text{ N}$ $W_B = 1000\text{ N}$
 $\mu = 0.25$ for all surfaces.
 Find inclination α for w
 the Block B impends sliding
 down the plane. String is
 || to the inclined plane find
 Tension in string also.



$$\Sigma F_x = 0 \text{ Gives}$$

$$T = \mu R_1 + 400 \sin \alpha$$

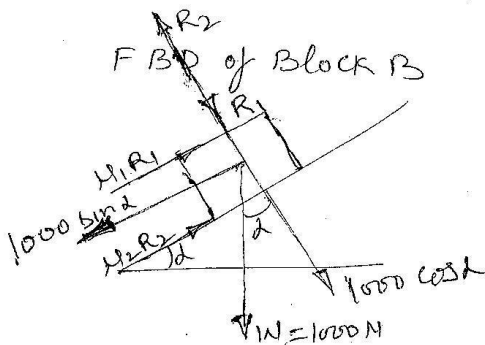
$$T = 0.25 R_1 + 400 \sin \alpha$$

$$\Sigma F_y = 0 \text{ Gives}$$

$$R_1 = 400 \cos \alpha \quad \text{--- (1)}$$

$$T = 0.25(400 \cos \alpha) + 400 \sin \alpha$$

$$T = 100 \cos \alpha + 400 \sin \alpha \quad \text{--- (2)}$$



$$\Sigma F_x = \mu R_1 + \mu R_2 = 1000 \sin \alpha$$

$$\mu (R_1 + R_2) = 1000 \sin \alpha$$

$$0.25 (R_1 + R_2) = 1000 \sin \alpha \quad \text{--- (3)}$$

$$\Sigma F_y \quad R_2 = 1000 \cos \alpha + R_1$$

$$R_2 = 1000 \cos \alpha + 400 \cos \alpha$$

$$R_2 = 1400 \cos \alpha \quad \text{--- (4)}$$

Now from Eqn (3)

$$0.25 (400 \cos \alpha + 1400 \cos \alpha) = 1000 \sin \alpha$$

$$= 100 \cos \alpha + 350 \cos \alpha = 1000 \sin \alpha$$

$$450 \cos \alpha = 1000 \sin \alpha$$

$$\frac{\sin \alpha}{\cos \alpha} = \frac{450}{1000} = 0.45$$

$$\tan \alpha = 0.45$$

$$\alpha = \tan^{-1}(0.45) = \underline{\underline{24.23^\circ}}$$

From (2)

$$T = 100 \cos 24.23 + 400 \sin 24.23 = \underline{\underline{255.35\text{ N}}}$$

